On rank-two norm-attaining operators on real Banach spaces

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Abstract

We study rank-two norm-attaining operators from real Banach spaces into the Euclidean plane. By perturbing zero coordinates in a weighted dual block, we construct a rank-k norm-attaining operator into ℓ_2^k for every integer $k \geq 2$ on the specific renorming of c_0 introduced by Martín. The case $k = 2$ answers his Open Problem 7 affirmatively. For a fixed norming pair, we characterize the existence of a mate by a local quadratic lower bound and by a closed absolutely convex hull in the kernel of the norming functional. We give a difference norm under which a rank-two norm-attaining operator and a norm-attaining functional without a mate coexist. Finally, a quadratic remainder estimate on the entire unit ball yields norm-attaining operators of any prescribed finite rank on another renorming of c_0.

Keywords norm-attaining operators, finite-rank operators, norming functionals, Banach space renorming, quadratic remainders

1 Introduction

Kadets, López, Martín, and Werner asked whether, for any two Banach spaces of dimension at least two, there exists a rank-two norm-attaining operator from the first into the second. They showed that, to establish this universal existence statement, it suffices to consider the Euclidean plane ℓ_2^2 as the range. They also proved existence whenever the set of norm-attaining functionals on the domain contains a nontrivial cone [1].

Martín constructed an equivalent renorming $\mathfrak{X}$ of c_0 whose set of norm-attaining functionals contains no nontrivial cone. In Open Problem 7 he asked whether $\mathfrak{X}$ nevertheless admits a rank-two norm-attaining operator into ℓ_2^2 [2, Theorem 1 and Open Problem 7]. Theorem 2 answers this question affirmatively and gives norm-attaining operators of any prescribed finite rank on the same space. The proof perturbs a norming functional along selected zero coordinates of a weighted dual block while controlling the norm of the resulting operator. Corollary 2.1 shows that $\mathfrak{X}$ also gives a negative answer to [1, Problem 13.12], which asks whether the nontrivial cone condition is necessary for the existence of rank-two norm-attaining operators into every range of dimension at least two.

The existence of a rank-two norm-attaining operator into ℓ_2^2 can also be formulated as the existence of a norm-attaining functional with a mate. Propositions 3 and 4 give two criteria for a fixed norming pair. Proposition 6 constructs, under a single equivalent norm, a rank-two norm-attaining operator and a particular norm-attaining functional without a mate. Proposition 7 derives a construction from a quadratic remainder estimate for a convex function defining the unit ball. Theorem 8 applies the construction to another renorming of c_0. The general existence question lies beyond the results established here.

2 Preliminaries

All spaces are over the real field. Let p denote the given norm on X, let B_p and S_p denote its closed unit ball and unit sphere, and let p^* denote its dual norm. A bounded linear operator $T : X \to H$ into a Hilbert space H attains its norm at $x_0 \in S_p$ if $\|Tx_0\| = \|T\|$.

A pair (x_0, f) is a *norming pair* if $x_0 \in X$, $f \in X^*$, and $p(x_0) = p^*(f) = f(x_0) = 1$. For any $f \in X^*$ with $p^*(f) = 1$, a *mate* for f is a functional $g \in X^* \setminus \{0\}$ satisfying

$$f(x)^2 + g(x)^2 \le p(x)^2 \qquad (x \in X). \tag{1}$$

The operator $T = (f, g)$ has norm one because (1) gives $\|T\| \le 1$ and $p^*(f) = 1$. If $g = \lambda f$, then $\|T\| = \sqrt{1 + \lambda^2}$, so $\lambda = 0$. Since $g \neq 0$, the operator has rank two. If (x_0, f) is a norming pair, (1) gives $g(x_0) = 0$, and T attains its norm at x_0 with $Tx_0 = (1, 0)$. Conversely, given a rank-two norm-attaining operator into ℓ_2^2, divide it by its norm and apply an orthogonal transformation to the range so that the image of a norm-attaining point is $(1, 0)$. The two components give a norm-attaining unit functional and a mate. By Euclidean duality, (1) is equivalent to

$$p^*(sf + tg) \le \sqrt{s^2 + t^2} \qquad (s, t \in \mathbb{R}).$$

We use this characterization from [1] to verify the constructions below.

Kadets, López, Martín, and Werner give the dual unit ball formula for the sum norm in [3, Lemma 2.1(a)]. In Proposition 1, we rederive the formula and give its equivalent minimum formulation.

Proposition 1. *Let q be a Banach space norm on X and let $R : (X, q) \to W$ be a bounded linear operator into a Banach space W. Set $p = q + \|R(\cdot)\|_W$. Then, for every $h \in X^*$,*

$$p^*(h) = \min_{h = u + R^* w} \max\{q^*(u), \|w\|_{W^*}\}.$$

Proof. Set $C = B_{q^*} + R^*(B_{W^*})$. The weak-star continuity of R^* makes C a weak-star compact, absolutely convex set. Taking the suprema over the two balls separately gives $\sup_{h \in C} h(x) = q(x) + \|Rx\|_W$ for every $x \in X$. The bipolar theorem [4, Theorem V.28] therefore gives $C = B_{p^*}$.

For any decomposition $h = u + R^* w$, put $M = \max\{q^*(u), \|w\|_{W^*}\}$. For every $x \in X$,

$$|h(x)| \le q^*(u) q(x) + \|w\|_{W^*} \|Rx\|_W \le M p(x).$$

Hence $p^*(h) \le M$. If $h \neq 0$, decompose $h/p^*(h)$ as an element of C and multiply the two terms by $p^*(h)$. This gives a decomposition that attains the minimum. For $h = 0$, take the zero decomposition. □

3 Main results

3.1 Finite-rank constructions on Martín's space

We describe the renorming of the real space c_0 used in Theorem 1 of [2]. In Steps 1 and 2, Martín equips c_0 with a smooth equivalent norm q whose norm-attaining functionals are

precisely the finitely supported elements of ℓ_1. He then chooses a dense operator range $Y \subset (c_0, q)^*$ such that $Y \cap c_{00} = \{0\}$ and an injective norm-one operator $J : \ell_1(\mathbb{N}\times\mathbb{N}) \to (c_0, q)^*$ with range contained in Y. If $e_{n,j}$ denotes the canonical vector at (n, j), set $v^*_{n,j} = J(e_{n,j})$. For each j, the normalized functionals $v^*_{n,j}/q^*(v^*_{n,j})$ are dense in the dual unit sphere. In particular, the family $(v^*_{n,j})_{n\geq 1}$ separates points of c_0, and its finite subfamilies are linearly independent by injectivity of J.

For $j, n \geq 1$, set $\Phi_j(n) = 2^{-j-n}q^*(v^*_{n,j}) > 0$. Let $E_j = (\ell_1, |\cdot|_j)$ be the renormed block of Lemma A in [2]. Its unit ball is $B_{\ell_1} + D_j(B_{\ell_2})$, where $D_j(a) = (\Phi_j(n)a_n)_{n\geq 1}$. Thus the dual norm on ℓ_∞ is

$$|a|^*_j = \|a\|_\infty + \|\Phi_j a\|_2.$$

Define

$$(R_j x)(n) = j2^{-j-n}v^*_{n,j}(x), \qquad W = \left(\bigoplus_{j\geq 1} E_j\right)_{\ell_1}, \qquad Rx = (R_j x)_{j\geq 1}.$$

Each R_j is injective because its coordinate functionals separate points. Also $|R_j x|_j \leq \|R_j x\|_1 \leq j2^{-j}q(x)$, so $R : (c_0, q) \to W$ is bounded. The space constructed by Martín is

$$\mathfrak{X} = (c_0, p), \qquad p(x) = q(x) + \|Rx\|_W = q(x) + \sum_{j\geq 1} |R_j x|_j.$$

Theorem 1 of [2] shows that the norm-attaining functionals in any two-dimensional subspace of $\mathfrak{X}^*$ are contained in at most two lines. Open Problem 7 in that paper asks whether this particular space admits a rank-two norm-attaining operator into ℓ_2^2.

Theorem 2. *Let $c_0(\mathbb{R})$ be the space of real sequences converging to zero. Take the base norm q, the coordinate functionals $v^*_{n,j}$, and the blocks $E_j = (\ell_1, |\cdot|_j)$ from the construction above. Define*

$$(R_j x)(n) = j2^{-j-n}v^*_{n,j}(x), \qquad p(x) = q(x) + \sum_{j\geq 1} |R_j x|_j,$$

and set $\mathfrak{X} = (c_0(\mathbb{R}), p)$. For each integer $k \geq 2$, there exist $x_0 \in S_p$ and a bounded linear operator $T : \mathfrak{X} \to \ell_2^k$ such that

$$\|T\| = 1, \qquad Tx_0 = (1, 0, \ldots, 0), \qquad \operatorname{rank} T = k.$$

In particular, Open Problem 7 of [2] has an affirmative answer for $\mathfrak{X}$.

Proof. Fix m and choose distinct indices $n_1, \ldots, n_{k-1}$. Set $b_i = R^*_m e_{n_i}$, where e_{n_i} denotes the coordinate functional on E_m. The functionals b_i are linearly independent. Since c_0 is infinite-dimensional, we can take a nonzero vector in $\bigcap_i \ker b_i$ and normalize it with respect to p to obtain x_0. Then $R_m x_0 \neq 0$ and $(R_m x_0)(n_i) = 0$ for every i.

Choose a functional u with $q^*(u) = 1$ and $u(x_0) = q(x_0)$. For each nonzero $R_j x_0$, choose a unit functional a_j that norms it in E_j. Set $a_j = 0$ if $R_j x_0 = 0$. Let $w = (a_j)_j \in W^*$ and $f = u + R^* w$. Since $\|w\| \leq 1$, Proposition 1 gives $p^*(f) \leq 1$, and

$$f(x_0) = q(x_0) + \sum_j |R_j x_0|_j = p(x_0) = 1.$$

Write $a = a_m$. If $a(n_i) \neq 0$, setting this coordinate to zero preserves the value at $R_m x_0$. Since $\Phi_m(n_i) > 0$, the weighted ℓ_2 norm strictly decreases and the supremum norm does not increase. This contradicts the fact that a is a unit norming functional for the nonzero vector $R_m x_0$. Hence $a(n_i) = 0$ for every i.

Set $A = \|a\|_\infty$ and $B = \|\Phi_m a\|_2$. Both are positive and $A + B = 1$. Choose $\beta > 0$ such that $\beta \leq A$ and $\beta \max_i \Phi_m(n_i) \leq B$. The disjointness of the supports gives, for $s \in \mathbb{R}$ and $t = (t_i) \in \mathbb{R}^{k-1}$,

$$\left| sa + \beta \sum_i t_i e_{n_i} \right|_m^* = \max\{|s|A, \beta\|t\|_\infty\} + \left(s^2 B^2 + \beta^2 \sum_i \Phi_m(n_i)^2 t_i^2 \right)^{1/2}$$
$$\leq (A + B)\sqrt{s^2 + \|t\|_2^2}. \tag{2}$$

Set $g_i = \beta b_i$. In the decomposition of $sf + \sum_i t_i g_i$, the component in $(X, q)^*$ is su and each unperturbed block is sa_j. Their norms are bounded by $|s|$, while (2) bounds the perturbed block. Proposition 1 thus gives

$$p^*\left(sf + \sum_i t_i g_i \right) \leq \sqrt{s^2 + \|t\|_2^2}.$$

Taking the supremum over the Euclidean unit ball shows that $T = (f, g_1, \ldots, g_{k-1})$ is a contraction. The g_i are linearly independent and vanish at x_0, whereas $f(x_0) = 1$. Hence T has rank exactly k and attains its norm at x_0. □

The project theorem registry records that its reconstruction of the rank-two proof drew on an external draft titled *A rank-two norm-attaining operator on Martín's renorming of* c_0 [5].

Corollary 2.1. *For every real Banach space Y of dimension at least two, there is a rank-two norm-attaining operator from $\mathfrak{X}$ into Y. Since the set of real norm-attaining functionals on $\mathfrak{X}$ contains no nontrivial cone, this gives a negative answer to [1, Problem 13.12].*

Proof. Take T and x_0 from Theorem 2 with $k = 2$. As in [1, Remark 13.2(a)], choose a norm-one isomorphism S from ℓ_2^2 onto a two-dimensional subspace of Y. The operator S attains its norm at some $y \in S_{\ell_2^2}$. Choose an orthogonal map Q with $QTx_0 = y$. Then SQT has norm at most one, attains the value $\|SQTx_0\| = 1$, and has rank two. The assertion about norm-attaining functionals follows from [2, Theorem 1]. □

3.2 Criteria for a fixed norming pair

Proposition 3. *Fix a norming pair (x_0, f) and put $E = \ker f$. The functional f has a mate if and only if there exist $\ell \in E^* \setminus \{0\}$ and $c, \varepsilon > 0$ such that*

$$p(x_0 + u) \geq 1 + c|\ell(u)|^2 \qquad (u \in E,\ p(u) \leq \varepsilon). \tag{3}$$

Proof. If g is a mate for f, then $\ell = g|_E \neq 0$ and $p(x_0 + u) \geq \sqrt{1 + |\ell(u)|^2}$. Since $\ell(u)$ is bounded for $p(u) \leq \varepsilon$, the identity $\sqrt{1 + r^2} - 1 = r^2/(\sqrt{1 + r^2} + 1)$ gives (3).

Conversely, set $Pz = z - f(z)x_0$. Given $z \in B_p$, by symmetry we may first assume that $f(z) = 1 - \delta \in [0, 1]$. Put $s = \varepsilon/(\varepsilon + 2)$. Then

$$(1 - s)x_0 + sz = (1 - s\delta)\left(x_0 + \frac{sPz}{1 - s\delta}\right) \in B_p.$$

Since $p(Pz) \le 2$, we have $p(sPz/(1 - s\delta)) \le 2s/(1 - s) = \varepsilon$. By (3),

$$\frac{cs^2|\ell(Pz)|^2}{1 - s\delta} \le s\delta, \qquad cs|\ell(Pz)|^2 \le \delta.$$

Define $g = \sqrt{cs}\,\ell P$. Then $f(z)^2 + g(z)^2 \le (1 - \delta)^2 + \delta \le 1$. For $f(z) < 0$, apply the same estimate to $-z$. Homogeneity gives (1), and $P|_E = I_E$ ensures that $g \neq 0$. □

The following corollary derives the dual quadratic criterion of Kadets, López, Martín, and Werner [1, Proposition 13.41] from Proposition 3.

Corollary 3.1. *Let $f \in S_{X^*}$. Then f has a mate in the sense of* (1) *if and only if there exist $h \in B_{X^*} \setminus \{0\}$ and $K, \delta > 0$ such that*

$$\|f + th\| \le 1 + Kt^2 \qquad (|t| < \delta).$$

Equivalently, for some such h, $\limsup_{t\to 0}(\|f + th\| - 1)/t^2 < \infty$. For every h satisfying the upper bound, there is $a \in (0, 1]$ such that ah is a mate for f.

Proof. If g is a mate, Euclidean duality gives $\|f + tg\| \le \sqrt{1 + t^2} \le 1 + t^2/2$. Conversely, suppose the upper bound holds for h. On X^{**} define $F(z) = z(f)$ and $H(z) = z(h)$. These are the canonical images of f and h in X^{***}, so $\|F + tH\| = \|f + th\|$. By Hahn–Banach, there is $z_0 \in S_{X^{**}}$ with $F(z_0) = 1$. Evaluating $F + tH$ at z_0 for both signs of t gives $H(z_0) = 0$.

For $u \in E = \ker F$ of sufficiently small norm, take $t = H(u)/(2K)$. Then $|t| < \delta$ and $|H(u)|^2/(4K) \le 1$, so

$$\|z_0 + u\| \ge \frac{1 + tH(u)}{1 + Kt^2} = 1 + \frac{|H(u)|^2/(4K)}{1 + |H(u)|^2/(4K)} \ge 1 + \frac{|H(u)|^2}{8K}.$$

Thus the hypothesis of Proposition 3 holds on X^{**} with $\ell = H|_E$. For the projection $Pz = z - F(z)z_0$ onto E, the equality $H(z_0) = 0$ gives $\ell P = H \neq 0$. The proof of Proposition 3 therefore produces a mate for F that is a positive multiple of H. Restricting to the canonical copy of X in X^{**} gives a mate for f of the form ah. We may decrease a if necessary to obtain $0 < a \le 1$. □

Proposition 4. *For the same norming pair, define*

$$\phi(u) = p(x_0 + u)^2 - 1, \qquad m(u) = \inf_{t\neq 0} \frac{\phi(tu)}{t^2}, \qquad K = \overline{\operatorname{aconv}}\{u \in E : m(u) \le 1\}.$$

Here $\overline{\operatorname{aconv}}$ denotes the norm-closed absolutely convex hull in E. The functional f has a mate if and only if $K \neq E$.

Proof. By (1), the restriction $\ell = g|_E$ of any mate satisfies $\ell^2 \le \phi$. Conversely, given a nonzero $\ell \in E^*$ with this property, define $g = \ell P$, where $Px = x - f(x)x_0$. Write $x = sx_0 + u$ with $u \in E$. When $s \neq 0$, applying the inequality at u/s and multiplying by s^2 gives (1). The case $s = 0$ follows by continuity.

Scaling along each line shows that $\ell^2 \le \phi$ is equivalent to $\ell^2 \le m$. The function m is nonnegative and homogeneous of degree two. Thus $\ell^2 \le m$ holds exactly when $|\ell|$ is bounded by one on $\{m \le 1\}$. Indeed, for $m(u) > 0$, apply the bound at $u/\sqrt{m(u)}$. If $m(u) = 0$, apply it to all scalar multiples of u. The bound extends to K, so the existence of a nonzero ℓ with this property implies $K \neq E$.

Conversely, letting $|t| \to \infty$ gives $m(u) \le p(u)^2$, so K contains the closed unit ball of E. If $K \neq E$, Hahn–Banach separation gives a nonzero continuous linear functional bounded above on K. Since K is symmetric, we can rescale the functional so that its absolute value is bounded by one on K. □

Example 5. On $\mathbb{R}^3$, set $p(x) = (\|x_+\|_2^4 + \|x_-\|_2^4)^{1/4}$, where x_+ and x_- are the coordinatewise positive and negative parts. To see that p is a norm, use the coordinatewise bounds $(x+y)_\pm \le x_\pm + y_\pm$, followed by the triangle inequalities in ℓ_2 and ℓ_4^2. Take $x_0 = (1,0,0)$ and $f(x) = x_1$, and write the vectors in E as (a,b). If $A = a_+^2 + b_+^2$ and $B = a_-^2 + b_-^2$, then for $t > 0$,

$$\frac{\phi(t(a,b))}{t^2} = \frac{\sqrt{(1+t^2A)^2 + t^4B^2} - 1}{t^2} \ge A,$$

with limit A as $t \downarrow 0$. For negative t, the roles of A and B are interchanged. It follows that $m(a,b) = \min\{A, B\}$. The function m vanishes on both coordinate axes but satisfies $m(1,-1) = 1$. Therefore every linear functional with $\ell^2 \le m$ must vanish, and f has no mate. This example also shows that m need not be convex, since $m(1,0) = m(0,-1) = 0$ and $m(1/2, -1/2) = 1/4$.

3.3 A difference norm on c_0

Proposition 6. *Let $\varepsilon > 0$ and define on c_0*

$$D(x) = \sup_{n \ge 1} |x_n - x_{n+1}|, \qquad p(x) = \varepsilon \|x\|_\infty + D(x).$$

Under this equivalent norm, there exist a rank-two norm-attaining operator into ℓ_2^2 and a norm-attaining unit functional without a mate. Each coordinate truncation P_N has operator norm $\|P_N\|_p = 1 + \varepsilon^{-1}$.

Proof. The inequalities $\varepsilon\|x\|_\infty \le p(x) \le (\varepsilon+2)\|x\|_\infty$ show that p is an equivalent Banach space norm on c_0.

Let $h_+ = (1+\varepsilon)e_2^* - e_3^*$ and $h_- = (1+\varepsilon)e_2^* - e_1^*$. Both satisfy $|h_\pm(x)| \le p(x)$ and take the value one at $x_0 = e_2/(1+\varepsilon) \in S_p$. Set $f_0 = (h_+ + h_-)/2$ and $g_0 = (h_+ - h_-)/2$. Then

$$f_0(x)^2 + g_0(x)^2 = \frac{h_+(x)^2 + h_-(x)^2}{2} \le p(x)^2.$$

Since $g_0 \neq 0$, (1) shows that (f_0, g_0) is a rank-two norm-attaining operator.

Now let $f = (1+\varepsilon)e_1^* - e_2^*$. Again $|f| \le p$. For $v_\theta = (1, \theta, \theta^2, \ldots)$ with $0 < \theta < 1$, we have $D(v_\theta) = 1 - \theta$, and hence $f(v_\theta) = p(v_\theta) = 1 + \varepsilon - \theta$. In particular, f is a norm-attaining unit functional. If g were a mate for f, (1) would give

$$0 = g(v_\theta) = \sum_{n \ge 1} g(e_n)\theta^{n-1}.$$

The series follows from convergence in the norm of c_0, and the coefficients $g(e_n)$ are bounded. Letting $\theta \downarrow 0$ yields $g(e_1) = 0$. Repeatedly dividing by the next power of θ and passing to the limit yields $g(e_n) = 0$ for every n. The density of the finitely supported sequences gives $g = 0$, a contradiction.

Finally, $D(P_N x) \le \max\{D(x), |x_N|\}$, so $p(P_N x) \le p(x) + \|x\|_\infty$. This gives the upper bound. For the reverse bound, take x whose first N coordinates are one and whose subsequent coordinates are $\theta, \theta^2, \ldots$. Then

$$\frac{p(P_N x)}{p(x)} = \frac{1+\varepsilon}{\varepsilon + 1 - \theta} \longrightarrow 1 + \varepsilon^{-1} \qquad (\theta \uparrow 1).$$

□

3.4 Quadratic remainders of convex defining functions

Proposition 7. *Let $F : X \to \mathbb{R}$ be continuous and convex, with $B_p = \{F \le 1\}$. Suppose $x_0 \in S_p$, $F(x_0) = 1$, and $L \in X^*$ is a supporting functional at x_0 in the sense that $F(z) \ge 1 + L(z - x_0)$ for all $z \in X$. Assume $D = L(x_0) > 0$ and that a bounded linear operator $U : X \to H$ into a Hilbert space satisfies*

$$F(z) - 1 - L(z - x_0) \ge \|U(z - x_0)\|^2 \qquad (z \in B_p), \tag{4}$$

Set $f = L/D$, $Pz = z - f(z)x_0$, and $C = \sqrt{D} + \|Ux_0\|$. Then $T = (f, UP/C) : X \to \mathbb{R} \oplus_2 H$ has norm one, attains its norm at x_0, and has range $\mathbb{R} \oplus U(\ker f)$. If $\dim U(\ker f) \ge k - 1$, orthogonally projecting the second component onto a $(k-1)$-dimensional subspace of $U(\ker f)$ gives a rank-k norm-attaining operator into ℓ_2^k.

Proof. For $z \in B_p$, the supporting inequality gives $L(z) \le D$. The symmetry of B_p gives $|L(z)| \le D$, so $p^*(f) = f(x_0) = 1$. For $z \in B_p$ with $f(z) \ge 0$, put $\delta = 1 - f(z)$. From (4) and $F(z) \le 1$ we obtain $\|U(z - x_0)\| \le \sqrt{D\delta}$. Hence

$$\|UPz\| \le \sqrt{D\delta} + \delta\|Ux_0\| \le C\sqrt{1 - f(z)^2}.$$

For $f(z) < 0$, applying this estimate to $-z$ gives the same bound. Thus T is a contraction and $Tx_0 = (1, 0)$. The decomposition $X = \mathbb{R}x_0 \oplus \ker f$ yields the asserted range. If $\dim U(\ker f) \ge k - 1$, choose a $(k-1)$-dimensional subspace of $U(\ker f)$ and project the second component onto it. This preserves both contractivity and norm attainment at x_0. □

Theorem 8. *Define on c_0*

$$F(x) = \sum_{n \ge 1} x_n^{2n} + \sum_{n \ge 1} 2^{-n}(x_n + x_{n+1})^4.$$

Let p_1 be the Minkowski functional of $\{F \le 1\}$. For every integer $k \ge 2$, there exists a rank-k norm-attaining operator from (c_0, p_1) into ℓ_2^k.

Proof. Every point of c_0 has a neighborhood in which the tail coordinates are uniformly bounded in absolute value by some $\rho < 1$. In that neighborhood, a geometric series and the convergent series $\sum_n 2n\rho^{2n-1}$ bound the tails of the first series and its first derivatives, respectively. On bounded sets, the second series and its first derivatives are bounded termwise by constant multiples of 2^{-n}. Thus F is continuously Fréchet differentiable, even, and convex. Since $F(0) = 0$, the sublevel set $\{F \leq 1\}$ contains a ball about zero for the supremum norm. The inequality $F(x) \leq 1$ implies $|x_n| \leq 1$ for every n. The sublevel set is therefore a bounded, closed, absolutely convex neighborhood of zero, and p_1 is an equivalent norm.

Let x_0 be the normalization of $e_1 + \cdots + e_k$ with respect to p_1. Continuity and $p_1(x_0) = 1$ give $F(x_0) = 1$. Set $L = DF(x_0)$. At least one power term has strictly positive radial derivative at x_0, and all remaining terms have nonnegative radial derivative. Thus $D = L(x_0) > 0$.

For $a \neq 0$ and $s \in \mathbb{R}$, the supporting-line inequality for the convex function $v \mapsto v^n$ on $[0, \infty)$ at $v = 1$ gives

$$s^{2n} - a^{2n} - 2na^{2n-1}(s - a) \geq na^{2n-2}(s - a)^2.$$

Indeed, substitute $v = (s/a)^2$ into that inequality. Applying this bound with $a = (x_0)_n$ for $1 \leq n \leq k$ and using the nonnegativity of the convex remainders of the other terms gives (4) for

$$Uz = \left(\sqrt{n}\,|(x_0)_n|^{n-1} z_n\right)_{n=1}^{k}$$

with $p = p_1$. The operator U has rank k. Its restriction to the codimension-one subspace $\ker f$ has rank at least $k - 1$. Proposition 7 gives the desired operator. □

Under the same norm, a rank-two operator can also attain its norm at the unit vector on the positive first coordinate axis. Let $x_0 = ae_1$, where $a > 0$ satisfies $a^2 + a^4/2 = 1$. Then

$$DF(x_0) = (2a + 2a^3)e_1^* + 2a^3 e_2^*.$$

The exact remainder of the term x_1^2 gives (4) for $U = e_1^*$. Since the second coordinate coefficient of $DF(x_0)$ is nonzero, $e_1^* P = e_1^* - af$ is nonzero. Proposition 7 therefore gives a rank-two operator attaining its norm at x_0.

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